

Mixtures

When substances mix but keep their own properties, you have a mixture — sometimes you can see the parts, sometimes you can't.

What a mixture is

When two or more substances are mixed and **each keeps its own properties** (no chemical reaction), the result is a **mixture**. The substances in it are its **components**.

Non-uniform (heterogeneous)	Uniform (homogeneous)
Components are visible / unevenly spread	Components evenly spread, can't be seen separately
Sprout salad, sand in water	Sugar in water, air

Alloys are uniform mixtures of metals: **stainless steel** (iron, nickel, chromium, carbon), **brass** (copper + zinc), **bronze** (copper + tin). Ancient India called such mixtures *Mishraloha*.

Air is a mixture

Air is a **uniform** mixture — mainly nitrogen (~78%), oxygen, argon, carbon dioxide and water vapour. **Lime water turns milky** in air, showing carbon dioxide is present. Suspended dust particles are pollutants, not a true part of air.

Key idea: in a mixture, components do **not** react and can usually be separated by physical methods.

Where students lose marks

Trap 1 — Uniform vs non-uniform. Uniform = evenly mixed, parts not visible (sugar solution, air). Non-uniform = parts visible/uneven (salad, sand in water).

Trap 2 — Components keep their properties. In a mixture there is no chemical reaction; each substance stays itself.

Trap 3 — Air is a mixture, not a compound. Its gases are simply mixed (uniformly), not chemically combined.

Lime water test: turns milky with carbon dioxide (forms calcium carbonate).

Model answers — write them like this

Example A. Classify as uniform or non-uniform: (i) sugar solution (ii) sand in water.

Step 1: Sugar solution — evenly mixed, parts not visible → uniform.

Step 2: Sand in water — parts visible → non-uniform.

∴ (i) uniform (ii) non-uniform.

Example B. How does lime water show air contains carbon dioxide?

Step 1: Lime water (calcium hydroxide) left in air turns milky.

Step 2: CO₂ reacts with it to form insoluble calcium carbonate.

∴ The milkiess proves carbon dioxide is present in air.

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